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# FULLY EXECUTED HUGGINGFACE TRANSFORMERS PIPELINE (FIXED SUMMARIZATION)
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import os
import sys

# 1. ENVIRONMENT COMPATIBILITY SETUP
print("📦 Installing required HuggingFace packages...")
!{sys.executable} -m pip install -q transformers torch

import torch
from transformers import pipeline

# --- 🧪 2. DISCOVERY PHASE: INITIALIZE PIPELINES ---
print("\n🚀 Initializing pre-trained HuggingFace Pipeline APIs...")
# Sentiment Pipeline (Using text-classification mapping)
sentiment_pipe = pipeline("text-classification")
# Text Generation Pipeline (Default: GPT-2 - Fully supported in your environment)
generator_pipe = pipeline("text-generation", model="gpt2")

print("✅ Pipelines initialized successfully.\n")

# --- 🧠 3. TECHNICAL PHASE: TASK IMPLEMENTATION ---

# --- TASK A: THE AUDIT (Sentiment Evaluation with Sarcasm Edge Case) ---
print("=== 📊 TASK A: THE SENTIMENT AUDIT RESULTS ===")
audit_sentences = [
    "I am absolutely thrilled with how quickly customer service resolved my problem.",
    "The package arrived completely shattered and two days late.",
    "Oh fantastic, another unexpected fee added to my monthly billing statement.",
    "The device works okay, but the battery life could definitely be better.",
    "I don't have any strong feelings about this update either way."
]

sentiment_results = sentiment_pipe(audit_sentences)
for sentence, res in zip(audit_sentences, sentiment_results):
    print(f"💬 Text : \"{sentence}\"")
    print(f"🏷️ Label: {res['label']} | 📈 Confidence: {res['score']:.4f}\n")

# --- TASK B: THE CREATOR (Constrained Domain-Specific Generation) ---
print("=== 📝 TASK B: TEXT GENERATION OUTPUT ===")
prompt = "Our technical support team can help you resolve account access issues."
gen_output = generator_pipe(prompt, max_length=50, num_return_sequences=1)

print(f"💡 Input Prompt: \"{prompt}\"")
print(f"🖨️ Generated Extension:\n\"{gen_output[0]['generated_text']}\"\n")

# --- TASK C: THE EXECUTIVE SUMMARY (Handled via Text Generation Prompt) ---
print("=== 📰 TASK C: SUMMARIZATION OUTPUT ===")
news_article = (
    "In an effort to scale operational efficiency, major customer support centers are embracing a rapid digital evolution. Artificial intelligence systems are being deployed to handle repetitive customer inquiries. Recent field studies indicate that integration of AI-driven solutions can lower average user queue wait times by up to forty percent. However, increased reliance on automated systems can introduce severe operational friction. As organizations navigate this transition, balancing automation with complex emotional context, situational nuance, and literal parsing remains a critical challenge."
)

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